

MANOJ SANAGAPALLI

Bhattiprolu, Andhra Pradesh

+91 8309478980

shvsmanoj2007@gmail.com

LinkedIn

GitHub

Portfolio

PROFESSIONAL SUMMARY

Second-year Computer Science student with strong programming skills in Python, Java, C, and C++. Hands-on experience in developing web applications using HTML, CSS, JavaScript, and APIs, with a solid foundation in Data Structures and Algorithms. Actively involved in hackathons and project-based learning, with exposure to machine learning and team leadership.

EDUCATION

- B.Tech in Computer Science and Engineering (AIML)**, VIT-AP University, Andhra Pradesh 2024 – Present
CGPA: 8.77 / 10.0
- Higher Secondary Education**, Krishna Reddy College, Vijayawada, Andhra Pradesh 2022 – 2024
Percentage: 95.1 / 100
- Secondary Education**, VSEM High School, Bhattiprolu, Andhra Pradesh 2022
Score: 568 / 600

PROJECTS

Cosmic Explorer (Live Demo)

Aug 2025

- Designed a 3D space mission simulator to visualize orbital mechanics and mission planning concepts.
- Implemented an Intercept Mission Engine with real-time trajectory and ΔV calculations based on physics principles.
- Developed custom APIs to handle mission datasets, intercept computations, and AI-assisted mission queries.
- Integrated an AI Mission Assistant to guide users with contextual mission insights and physics explanations.
- Implemented IndexedDB for persistent local data storage, enabling offline mission history tracking.
- Combined frontend engineering, physics-based logic, API design, AI integration within a hackathon timeline.

Health Risk Score Predictor

Jan 2026

- Designed and implemented an end-to-end machine learning pipeline to predict a continuous health risk score from real-world health indicators.
- Performed data preprocessing, feature engineering, and standardization to prepare inputs for regression modeling.
- Implemented and compared Linear, Ridge, and Lasso regression models using RMSE and R^2 to evaluate accuracy and generalization.
- Analyzed coefficient behavior to understand feature importance and applied regularization to improve model stability and interpretability.

TECHNICAL SKILLS

- Programming Languages:** Python, C, C++, Java, SQL, HTML, CSS, JavaScript
- Frameworks:** NumPy, Pandas, Scikit-Learn, Matplotlib
- Tools:** Excel, PowerPoint, Tableau, MySQL
- Platforms:** PyCharm, Jupyter Notebook, Visual Studio Code, IntelliJ IDEA
- Core CS Concepts:** OOPs, DSA, Machine Learning
- Developer Tools:** Git, GitHub and Version Control
- Soft Skills:** Problem Solving, Effective Time Management, Leadership, Communication
- Languages known:** English, Telugu, Hindi

ACHIEVEMENTS

- National Finalist** – CTRL+SPACE Hackathon, Newton School of Technology (Team VOID ranked Top 5 nationwide)

POSITIONS OF RESPONSIBILITY

- Team Lead – Hackathon:** Led frontend development and AI integration, managed GitHub repositories, and coordinated project delivery.
- Marketing Team Lead** – Machine Learning Club, VIT-AP
- Marketing Head** – TEDxVITAP 2025
- Campus Ambassador** – ANEXZ Solutions

CERTIFICATIONS

- Introduction To Python - Microsoft
- TCS-iON Career Edge - Young Professional Certificate
- Programming in Java - Infosys
- AI Foundations Associate - Oracle